

IoT Meets Digital Twins

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Abstract—Critical systems such as industrial infrastructures and cyber-physical systems operate under strict constraints, making direct testing and modification of live systems highly risky. Digital Twin technologies, when integrated with the Internet of Things (IoT), enable virtual modeling, monitoring, and evaluation of such systems without disrupting real-world operations. This paper investigates the integration of Digital Twins and IoT, focusing on their conceptual evolution, practical applications, and deployment feasibility. The state of the art highlights the transition from static models to dynamic, data-driven, and AI-assisted digital twins. The readiness of these technologies is assessed based on factors including technological maturity, data quality, scalability, and adoption challenges. The findings show that, despite significant benefits, deployment readiness remains application-dependent and continues to evolve.

I. INTRODUCTION

A. Critical systems and lack of test environments

Consider an attempt to improve the efficiency of a power grid that fails due to unforeseen technical issues. Such a failure can disrupt not only electricity generation but also other dependent operations, potentially resulting in large-scale power outages across an area. Due to the strong inter-dependencies and important nature of these systems, directly introducing untested changes into a live, complex critical system is strongly discouraged and poses significant operational and safety risks. In highly interconnected systems, isolating a fault to a single component is often difficult, making even minor malfunctions potentially significant. Such errors can spread across dependent subsystems, leading to consequences such as energy inefficiencies, operational delays, resource wastage, and substantial economic losses. Furthermore, critical systems typically operate under strict regulatory frameworks, service-level agreements, and standard operating procedures, which severely limit the feasibility of direct modifications to live infrastructures. Predicting the exact impact of introducing a new change in complex systems with multiple interdependent subsystems and partially unknown dynamics is, therefore, highly challenging and troublesome. Hence, alternative approaches are required to evaluate and validate system changes without disrupting real-world operations. [Faizan Bhagat]

B. Role of IoT in modern cyber-physical systems

Cyber-physical systems (CPS) refer to a fusion of physical processes with computing and communication. One of the most important components of CPS is the Internet of Things (IoT), which facilitates continuous sensing, data transmission

in real-time, and distributed intelligence. Sensors, actuators, and embedded system components of IoT are responsible for sense data related to physical assets to be processed in cyber space [1].

By means of standardized communication protocols and network infrastructures that can scale to accommodate a multitude of devices and platforms, the IoT makes interoperability between heterogeneous devices and platforms possible. The interoperability made possible by the IoT is critical for the large-scale deployment of CPSs in areas such as the energy domain and industrial automation. Furthermore, the IoT makes edge cloud computing technologies necessary for latency-sensitive applications [2].

In essence, IoT represents the roots of the CPS, enabling the connection between the physical and digital realms, ensuring the capability for real-time observation, and facilitating intelligent and autonomous operations. [Abdul Rauf]

C. Digital twins as virtual representations

Digital twins are commonly defined as virtual representations of physical assets, processes, or systems that are dynamically linked to their real-world counterparts through continuous data exchange. Recent surveys emphasize that, unlike traditional simulation models that are typically static and executed offline, digital twins are designed to remain synchronized with the physical system throughout its operational lifecycle by continuously incorporating real-time and historical data streams [3] [4]. This continuous synchronization enables digital twins to capture not only nominal system behavior but also performance degradation, environmental influences, and evolving usage patterns.

Contemporary digital twin definitions extend beyond early conceptual models by explicitly integrating lifecycle awareness and operational feedback. Recent literature describes digital twins as executable digital artifacts that evolve alongside the physical system, supporting monitoring, analysis, and prediction under real operating conditions [4] [5]. In cyber-physical systems, this tight coupling allows the digital twin to serve as an operational model capable of reflecting system dynamics with sufficient fidelity to support decision-making and system optimization.

The Internet of Things (IoT) plays a fundamental role in enabling this persistent synchronization. Modern digital twin frameworks rely on distributed IoT sensors and actuators embedded within physical systems to provide continuous telemetry, including state measurements, environmental

parameters, and operational events [3] [6]. These data streams are transmitted through communication middleware and integrated into the digital twin, enabling near real-time updating of the virtual model. As highlighted in recent IoT-focused surveys, this data-driven coupling allows digital twins to move beyond visualization and act as analytical entities capable of diagnosing system behavior and evaluating operational conditions [6].

In IoT-enabled cyber-physical systems, digital twins therefore function as operational representations rather than purely descriptive models. By integrating sensing, data processing, and system logic into a unified framework, digital twins create an abstraction layer between raw IoT data and higher-level control, optimization, or decision-support functions [3] [4]. This capability is particularly valuable in complex or safety-critical infrastructures, where experimentation on live systems is infeasible, and system behavior must be assessed through virtual yet continuously synchronized representations. [Rilwan]

II. STATE OF THE ART: DIGITAL TWINS AND IOT SYSTEMS

A. Evolution of Digital Twin Concepts

In earlier stages, engineers relied on basic models of critical systems that were primarily developed during the system design phase. While these models were useful at the time of design, they were not suitable for real-time operation or for accurately simulating physical systems when system data varied and was continuously updated [7]. As a result, such models quickly lost relevance once the physical system entered live operation.

With further technological evolution, the emergence of the Internet of Things (IoT) marked a major breakthrough. IoT enabled engineers to acquire real-time data from physical systems through sensors measuring parameters such as temperature, pressure, voltage, vibration, and other usage patterns. The availability of continuous real-world data allowed system models to be combined with live sensor information, leading to the development of digital twins. Digital twins can be described as software-based virtual representations of physical systems that are continuously updated using real-world data obtained through IoT technologies. To further understand the concept of a digital twin, it can be decomposed into three mandatory components. The first component is the physical system itself, such as a machine, production line, or power grid. The second component is the digital model, which is a software-based representation designed to imitate the behavior and characteristics of the physical system. The third component is the data connection layer, which consists of IoT sensors and communication technologies that enable continuous data exchange between the physical critical system and its corresponding digital twin [7], [8].

With time, the integration of Artificial Intelligence (AI) and Machine Learning (ML) techniques significantly advanced the capabilities of digital twins, taking them beyond simple real-time mirroring. These techniques enabled digital

twins to learn patterns from historical and real-time data, predict system behavior and potential failures, detect anomalies, and adapt to changing operating conditions [9], [10]. Consequently, the early static representations of physical systems evolved into intelligent and adaptive digital twin systems capable of supporting advanced monitoring, prediction, and decision-making processes. [Faizan Bhagat]

B. IoT Architectures Supporting Digital Twins

Digital twin systems use the architecture of the Internet of Things to ensure that the virtual twin runs in synchronization with the actual system. The architecture used includes the sensing components, communications, processing layers, and application services. The physical system has sensors that continuously acquire the information that can be communicated using the lightweight communication protocols such as MQTT/HTTP protocols [11].

Processing units for edge and cloud computing involve data filtering, aggregation, and analysis. Scalability, low latency, and guaranteed delivery of data to digital models are ensured in these units. Time series databases and microservices are usually used to handle data from sensors at a higher frequency [2].

In the application layer, the digital twin changes its internal state in real time, thus enabling visualization, simulation, and control. The two-way communication process makes it possible for control commands from the digital twin to be implemented on the physical system. Therefore, the architectures of IoT are very precious for realizing a digital twin implementation, being precise, real time, and scalable. [Abdul Rauf]

C. AI-powered and data-driven digital twins

The growing complexity of cyber-physical systems, together with the high volume and heterogeneity of data produced by IoT infrastructures, has accelerated a shift from purely physics-based digital twin models toward AI-powered and data-driven approaches. Recent surveys identify this transition as a major trend in the current state of the art, particularly for systems characterized by nonlinear behavior, uncertainty, or incomplete physical knowledge [3] [4].

AI-powered digital twins exploit machine learning techniques to infer system behavior directly from historical and real-time IoT data, reducing reliance on manually derived mathematical models. As discussed by Sala et al. and Shao et al., data-driven modeling allows digital twins to continuously adapt and evolve around operating conditions, allowing them suitable for deployment in dynamic environments such as smart manufacturing [3] [4]. Commonly adopted techniques include regression-based models, deep neural networks, and ensemble learning methods, which are used to approximate system dynamics, forecast future states, and identify anomalous behavior.

Recent industrial-oriented studies further highlight the practical relevance of AI-based digital twins in predictive maintenance and condition monitoring scenarios. The surveys

in [5] [6] report multiple implementations where machine learning-driven twins are trained using operational IoT data to estimate degradation trends, detect early faults, and support maintenance decision-making. These approaches are particularly effective in contexts where physical degradation mechanisms are difficult to model explicitly or vary across operational regimes.

At the same time, contemporary literature emphasizes that AI-powered digital twins introduce new technical challenges that directly affect their deployment readiness. Data quality and representativeness strongly influence model accuracy, while the black-box nature of many machine learning models limits interpretability and complicates validation in safety-critical applications [5]. In addition, the computational demands associated with continuous learning and real-time inference raise scalability concerns, especially for large-scale IoT systems [6]. These limitations motivate current research efforts toward hybrid digital twins that combine data-driven learning with simplified physical constraints, as well as distributed and edge-based architectures, which are further discussed in the following sections. [Rilwan]

III. DIGITAL TWIN EXAMPLES IN IOT SYSTEMS

A. Industrial IoT and smart manufacturing digital twins

Based on the architectural foundations discussed above, the following section reviews real-world implementations. Digital twins integrated with the Internet of Things (IoT) in industrial environments can optimize operations to a significant extent. One prominent example is the enhancement of predictive maintenance for industrial machines. Traditionally, maintenance of production machinery is performed either after a breakdown occurs or based on fixed schedules. Both approaches are often inefficient, as breakdown-based maintenance leads to unplanned downtime, while regular scheduled maintenance can be time-consuming and unnecessary in certain cases.

With the advancement of industrial IoT, sensors are integrated directly into physical systems, enabling continuous collection of operational data such as vibration, temperature, speed, and pressure. This data is continuously monitored and compared with historical data representing normal machine behavior. When deviations or abnormal patterns are detected in the recent data, the digital twin can predict potential failures before they occur, allowing proactive maintenance to be scheduled accordingly. As a result, maintenance strategies can be evaluated and optimized without directly interrupting the real physical production line. Furthermore, analysis assisted by Artificial Intelligence can enhance fault detection accuracy and improve the predictive capabilities of industrial digital twins [9], [10].

In addition to predictive maintenance, the use of digital twins with industrial IoT can also optimize production processes. Multiple machines within a single production line can be interconnected through a virtual twin, enabling continuous monitoring of production data. This allows inefficiencies in production processes or deviations from normal operating

conditions to be identified digitally. Consequently, production parameters can be tested and optimized virtually before being applied to the physical system. Moreover, factory operators benefit from a centralized view of the entire production line, improving operational visibility and decision-making.

[Faizan Bhagat]

B. Smart grid and infrastructure digital twins

While digital twins have shown strong potential in industrial manufacturing environments, similar concepts are increasingly being applied in large-scale infrastructure systems such as smart grids. Digital twins for smart grids use IoT technology to simulate and manage electricity generation, distribution, and supply chains. Various sensors are spread throughout the grid to provide information about voltage, current, power, and grid component status. All this data helps to maintain updated models of the grids. Data analysis is done in real-time [12].

Digital twins help in simulating fault situations, load optimization, and analysis of stability without interrupting process operations. Likewise, Infrastructure Digital Twins for buildings, bridges, and transport networks use IoT-based structural and environmental sensing data for condition-based maintenance and management [13].

Smart grid and infrastructure digital twins allow proactive decision-making and risk-free simulation, enhancing system reliability, efficiency, and safety. [Abdul Rauf]

C. Machine learning-based and black-box digital twins

Various studies have shown the partial or complete implementation of machine learning-based digital twins in real-world IoT systems. A representative example is the work by Carotenuto et al., who implemented an online black-box digital twin for industrial IoT systems using neural networks trained on live sensor data [9]. Their approach focused on modeling system behavior without explicit physical equations, enabling deployment in environments where system dynamics are partially unknown.

In this implementation, IoT sensors continuously streamed operational data to a learning module, which updated the digital twin in real time. The black-box twin was used to predict system outputs and detect anomalies during operation, demonstrating feasibility in industrial monitoring scenarios. Importantly, the authors highlighted that the digital twin was not a standalone simulator but an embedded component of the operational pipeline, tightly integrated with IoT data flows.

Similar approaches have been applied in energy systems, where machine learning-based digital twins are used to model grid components and predict load behavior. Ye et al. reported a black-box digital twin for power transformers using deep learning models trained on historical and operational IoT data, enabling fault detection without explicit electromagnetic modeling [14]. These examples show that black-box digital twins are already deployed in practice.

While these implementations demonstrate strong predictive performance, they also expose inherent limitations. The learned models are difficult to interpret, and their validity

outside the training data distribution is uncertain. These characteristics directly influence deployment readiness and motivate the evaluation of advantages and limitations. [Rilwan]

IV. READINESS, ADVANTAGES, DISADVANTAGES, AND OPEN ISSUES

A. Readiness levels and maturity assessment

The use-cases presented in the previous section explain the practical feasibility of digital twin technologies. However, their real-world deployment also raises important questions regarding readiness, scalability, and reliability, which are discussed in the following section. Despite the rapid emergence of IoT technologies and virtual twin concepts, the readiness of digital twin solutions for real-world industrial deployment remains an important concern [7]. The readiness of digital twins and industrial IoT systems can be evaluated across multiple dimensions. In terms of technological maturity, many industrial digital twin solutions are currently partially deployed and application-specific, while fully autonomous and large-scale digital twins are still under active development. As a result, the overall maturity level can be considered moderate. The operation and reliability of digital twins depend heavily on the quality and availability of data obtained from IoT sensors. In real-world industrial environments, sensor failures, noisy measurements, and incomplete data are common challenges [9]. Therefore, the presence of a robust and high-quality sensor infrastructure is a critical factor influencing deployment readiness. In addition, computational readiness and scalability play a significant role in determining the feasibility of digital twin implementations. Real-time digital twins require high computational resources and low-latency communication, and scaling from a small number of computing units to large, distributed industrial systems remains challenging [7].

Furthermore, the initial setup and deployment of digital twin systems often involve high costs, particularly in terms of sensor installation, data infrastructure, and computational resources. The lack of standardized validation frameworks further contributes to trust and reliability concerns, especially in safety-critical industrial applications [9]. Consequently, the adoption of digital twins and IoT technologies is currently selective, with implementations primarily observed in Industry 4.0 environments and pilot projects rather than universal industrial deployment [10]. [Faizan Bhagat]

B. Advantages of digital twin-IoT integration

The use of digital twins and IoT integration has several benefits, some of which include predictive maintenance and optimization. The continuous data flow from the IoT helps to keep the digital twins up to date regarding the actual system [11]. Based on the smart grid digital twin instance in Section III.B, one major benefit is enhanced real-time monitoring of grid conditions, which allows operators to detect faults earlier and respond quicker. In addition to this, virtual assessments regarding the operations improve the safety of the system by avoiding risks associated with the human and hardware components [1]. Control strategies can be tested virtually

on the digital twin before being applied to the real grid, which significantly reduces the risk of large-scale outages. The load prediction and energy distribution optimization discussed in Section III.B elaborate how digital twins aid improved decision-making and operational efficiency in smart grid systems. The merging of the systems enhances scalability and flexibility for the evolutionary development of the system that involves the expansion of the system through the addition of new sensors and analytics modules. [Abdul Rauf]

C. Limitations and technical challenges of machine learning-based digital twins

The practical implementations discussed in Section 3.3 reveal several limitations that directly affect deployment readiness. In machine learning-based industrial digital twins, data dependency emerges as a primary challenge. Carotenuto et al. reported that model accuracy degrades significantly under sensor faults or missing data, highlighting the vulnerability of black-box twins to imperfect IoT infrastructures [9]. This limits their reliability in harsh industrial environments.

Computational scalability is another critical issue. Online learning and real-time inference require continuous processing of high-frequency IoT data streams. In the transformer digital twin presented by Ye et al., centralized processing introduced latency constraints, suggesting that large-scale deployment would require edge-based or distributed architectures [14]. These constraints currently restrict black-box digital twins to subsystem-level applications rather than full system replication.

Model interpretability further limits adoption in safety-critical systems. Both studies acknowledge that black-box predictions are difficult to validate against physical principles, complicating certification and regulatory acceptance. This challenge motivates the ongoing shift toward hybrid digital twins that combine machine learning with simplified physics-based constraints to improve trustworthiness.

Finally, integration complexity remains an open issue. Implemented digital twins are often tightly coupled to specific sensor setups and data pipelines, reducing portability across platforms. This lack of standardization affects long-term maintainability and cross-domain reuse. Overall, while implemented machine learning-based digital twins demonstrate clear advantages in adaptability and predictive capability, their readiness remains application-specific and constrained by data quality, computational resources, and validation requirements. [Rilwan]

V. CONCLUSION

A. Summary of findings

This research paper investigated the integration of Digital Twin technologies with the Internet of Things, focusing on their necessity, applications, and readiness in real-world industrial environments. As critical systems cannot be directly interrupted for testing or evolution without risking operational, resource, and financial losses, the adoption of digital twins

integrated with IoT emerges as a necessary approach for safe technological advancement.

Initially, digital twins existed as static, design-time models with limited applicability during system operation. With the integration of IoT technologies, they evolved into dynamic, real-time, data-driven, and adaptive virtual representations. In industrial contexts, digital twins enable the detection of abnormal operational patterns, support predictive maintenance scheduling, and provide centralized visibility and control over production lines.

Although the adoption of digital twin and IoT solutions is currently moderate and largely limited to Industry 4.0 environments and pilot deployments, these technologies continue to scale and mature. As a result, digital twins are increasingly recognized as a foundational technology for the future development of intelligent and resilient industrial systems. [Faizan Bhagat]

B. Impact of IoT integration

Integration of the IoT is imperative in the development of dynamic digital twins from static models of simulation. The continuous flow of data between the physical and virtual twins is vital, allowing both to be synchronized at all times, hence enabling automation and analytics.

Moreover, the data accessible through the IoT can be utilized by machine learning algorithms and anomaly resolution algorithms to enable the growth of self-optimizing systems in digital twins. Thus, the integration of the IoT makes the applications of digital twins much more beneficial and sustainable in CPSs. [Abdul Rauf]

C. Role of AI in future digital twins

Future digital twin systems will evolve from static representations of physical assets to those of system companions with adaptive and autonomous characteristics. Support for learning, prediction, and optimization in the entire lifecycle of cyber-physical systems will be inextricably connected to and supported by artificial intelligence, which will play a central role in enabling this evolution. One potential path is the further use of AI for real-time decision support. Machine-learning-based models allow digital twins to adapt to new operational conditions on-the-fly, leading to proactive maintenance and fault anticipation and optimization, and it can be conducted without direct system testing in practice. Such a feature is especially useful in critical infrastructures because safety, reliability and regulatory compliance constraints do not permit direct modifying systems. In addition to the mentioned trend of virtualization of IoT infrastructures within digital twin environments. The modeling of sensing, communication, and data processing behavior next to physical processes allows for system-level characterization of robustness, latency, and resilience. This holistic view enables digital twins to evaluate not just how the physical system works, but the reliability of the data acquisition and communication layers that work to maintain performance. In summary, AI-powered digital twins are anticipated to mature into multi-faceted virtual

environments enabling safe experimentation, system co-design and autonomous optimization. And yet issues concerning scalability, validation, and trust persist, AI-enhanced digital twins represent an important enabling technology for the future creation of intelligent and resilient cyber-physical systems. [Rilwan]

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